OUSSAMA RHAMMI

Software Engineer · Full-Stack Developer · Systems Analyst | **TN Visa Eligible — Canadian Citizen**

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✓ No sponsorship required · No lottery · TN status obtained at the border on day of hire

PROFESSIONAL SUMMARY

Full-stack software engineer and systems analyst with 3 internships spanning game engine development, accessible software, and enterprise IT — including production C++ work at Ubisoft Montreal on Anvil/Scimitar (Rainbow Six Siege, Skull and Bones). Currently building a live multi-tenant SaaS product (Next.js, FastAPI, PostgreSQL, OpenAI API). Computer Science degree with Cybersecurity specialization from University of Sherbrooke, graduating May 2026. Authorized to work in the United States immediately under TN Visa — no sponsorship, no lottery required.

PROFESSIONAL EXPERIENCE

Software Engineer Intern — UI Engine Programmer

May 2025 – Aug 2025

Ubisoft Montreal · Montreal, QC

C++ · Anvil/Scimitar Engine · Game Pipeline · Performance Optimization · Regression Testing · CI/CD

- Shipped production C++ features inside Anvil/Scimitar, Ubisoft's proprietary engine powering Rainbow Six Siege and Skull and Bones — deployed across 30+ countries.
- Built an automated error-detection and reporting tool in C++ that reduced bug triage time for 200+ developers spanning international studios across North America and Europe.
- Developed performance optimization utilities and conducted functional/regression testing to maintain stability on active AAA production titles with millions of active players.
- Contributed to a large-scale shared codebase inside a 20,000+ employee global organization, collaborating daily with engineering, art, and QA departments.

Software Development Intern

Jan 2024 – Apr 2024

HumanWare · Montreal, QC

C# · .NET · MVVM · OCR · Image Processing · Cross-Platform · Agile/Scrum

- Built accessibility software for users with low vision in C# using MVVM architecture, integrating camera systems, OCR, and real-time image processing on constrained hardware.
- Delivered cross-platform features end-to-end within an Agile Scrum team, consistently meeting sprint commitments and contributing to code reviews.

IT Project Coordination Intern

Sep 2024 – Dec 2024

CIUSSS / CHUS — University Hospital · Sherbrooke, QC

Project Management · Enterprise IT · Requirements Analysis · Healthcare Compliance · Vendor Management

- Coordinated full IT project lifecycle for a 20,000+ employee healthcare organization — managed vendor communication, resource allocation, timelines, and compliance documentation.
- Produced technical analyses and requirements documentation; supported cross-functional teams across clinical and IT departments, gaining direct exposure to enterprise-scale governance.

PROJECTS

Ask Your Handbook — Multi-Tenant RAG SaaS

2025 – Present

Live Product

Next.js · FastAPI · PostgreSQL · OpenAI API (GPT-4o-mini) · SQLAlchemy · Docker · REST APIs · Resend · RBAC

- Architecting a production SaaS platform that enables SMBs to query internal documents via AI — multi-tenant with organization management, subscription tiers, and role-based access control (RBAC).
- Implemented complete onboarding: multi-step flow, email invitations via Resend, better-auth organization plugin, and freemium model with per-tenant document/member limits enforced at the API layer.
- Integrated OpenAI GPT-4o-mini + text-embedding-3-small for RAG retrieval; migrated from local Ollama/HuggingFace to a centralized API to serve non-technical business clients at scale.

GRCB Laboratory Management Platform — Capstone

2024 – 2025

University of Sherbrooke · Sherbrooke, QC

Java 17 · Quarkus · Angular · PostgreSQL · Keycloak · WebSocket · SSO · RBAC

- Designed and built a full-stack enterprise application replacing manual workflows for lab equipment scheduling and inventory at UdeS's Cement & Concrete Research Group.
- Architected a normalized PostgreSQL schema managing complex relationships across equipment, reservations, materials, and users; implemented real-time calendar sync via WebSocket.
- Integrated Keycloak with @usherbrooke.ca SSO and a granular RBAC system covering researchers, lab managers, and administrators.

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, JavaScript, TypeScript, SQL, Bash

Frontend: React, Next.js, Angular, Flutter, HTML/CSS, TypeScript

Backend & APIs: FastAPI, Node.js, Quarkus, REST APIs, WebSocket, .NET, Spring Boot

Databases: PostgreSQL, MySQL, Oracle, Firebase, MongoDB, SQLAlchemy

DevOps & Cloud: Docker, Git, CI/CD, Linux/Unix, Windows Server, Agile/Scrum, Jira

AI / ML: RAG pipelines, OpenAI API, LLM integration, TensorFlow, vector search, embeddings

Cybersecurity: Active Directory, DNS/GPO, Cryptography, Penetration Testing, Network Defense, SSO/Keycloak

Analysis & Design: Requirements analysis, technical documentation, UML, system design, enterprise IT coordination

Languages (Human): French — Native | English — Fluent

EDUCATION

B.Sc. Computer Science — Specialization: Cybersecurity & AI *Expected May 2026*

University of Sherbrooke · Sherbrooke, QC

Advanced coursework in AI, Software Engineering, Cybersecurity, Computer Graphics, and Distributed Systems

Diploma of College Studies — Computer Science Programming *2020 – 2023*

Collège de Montréal · Montreal, QC